MANASVA DHAKA

Software Engineer | B.Tech Student

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ABOUT ME

B.Tech Information Technology undergraduate at Manipal Institute of Technology with internship experience shipping a production-style full-stack application. Hands-on with REST API design, JWT authentication and role-based access control, MongoDB schema and aggregation design, real-time features with Socket.IO, and automated testing. Strong foundation in Java, C and Python with daily practice in data structures and algorithms, and a growing focus on backend systems and scalable software design. Looking to contribute as a Software Engineering Intern.

EXPERIENCE

Software Development Intern — Adnate IT Solutions Pvt Ltd (Remote) *Jun 2026 – Jul 2026*

- Built a full-stack Student Management System (React, Redux Toolkit, React Router; Node.js, Express, MongoDB) with full CRUD, search, filtering, sorting and pagination.
- Implemented JWT-based authentication with role-based access control (RBAC) and an OTP-driven forgot-password flow with automated email notifications.
- Built authenticated real-time features with Socket.IO, including live user-presence tracking and an admin activity feed that updates on every data change.
- Designed MongoDB aggregation pipelines to power dashboard analytics, including branch-wise distribution, average CGPA and registration trends.
- Added Excel, CSV and PDF export endpoints and implemented audit logging for all administrative actions.
- Wrote unit and integration tests (Jest, Supertest, mongodb-memory-server) covering authentication and core CRUD functionality.

PROJECTS

DistributedFS: Distributed File Storage System GitHub *Java · Distributed Systems · REST API · CI*

- Built a GFS/HDFS-inspired distributed file store in plain Java: a REST API server streams files in 1 MiB chunks to 3 storage-node processes over HTTP, verified by per-chunk and whole-file SHA-256.
- Implemented 2x replication with verified-read fallback, heartbeat failure detection (~5 s) and an on-demand repair that restores 2 verified copies; files stay readable with a node killed.
- Designed crash-safe uploads with a write-ahead pending marker and an atomic manifest commit; startup recovery through the node API finishes or rolls back interrupted uploads, tested by killing a JVM mid-upload.
- Wrote 629 tests and 10 stress tests (100 MiB in a 16 MiB heap), validated by mutation testing and run in GitHub Actions CI; benchmarked 63 MiB/s verified downloads.

AI-Powered Customer Support Chat Widget *FastAPI · Python · Gemini API · JavaScript*

- Built an embeddable AI chat widget that answers customer questions strictly from a business's own FAQ content, preventing hallucinated responses.
- Designed a FastAPI backend that grounds every response in a configurable knowledge base and keeps API keys server-side, shipped as a dependency-free widget that embeds with two script tags.

Personal Portfolio Website *HTML · CSS · JavaScript*

- Responsive portfolio showcasing projects and skills, iterated on UI/UX and accessibility from review feedback.

SKILLS

Languages: Java, C, Python, JavaScript
Web & Backend: React.js, Redux Toolkit, Node.js, Express.js, REST API Design, JWT Authentication & RBAC, Socket.IO
Databases: MongoDB, SQL
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Distributed Systems Fundamentals
Tools & Testing: Git & GitHub, GitHub Actions (CI), Jest, Supertest

EDUCATION

Manipal Institute of Technology, Bengaluru *2024 – 2028*

Bachelor of Technology (B.Tech), Information Technology • **CGPA: 7.34**

BVB Vidyashram, JLN Marg, Jaipur, Rajasthan

Class 12th: 88.2% | Class 10th: 89%

LANGUAGES

English, Hindi